

Nathaniel Mueller

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EDUCATION

Carnegie Mellon University, Pittsburgh, PA

May 2026

B.S. in Mathematical Sciences with concentration in Computational and Applied Mathematics

Cumulative QPA: 3.51, graduated with University honors

Selected Coursework: Differential Equations (Ordinary and Partial), Math Finance (I and II), Operations Research (I and II), Discrete Math, Probability, Environmental Economics, Machine Learning, Principles of Imperative Computation

DIS Study Abroad, Copenhagen, Denmark

Spring 2025

Core course in Artificial Neural Networks and Deep Learning; field studies at tech start-ups in Copenhagen and Berlin

SKILLS

Programming: Python, C/C++, SQL, Java, Matlab, R, Arduino

Libraries and Frameworks: NumPy, PyTorch, TensorFlow, Pandas, Keras, PySpark, Matplotlib, Databricks, Jupyter Notebook

Tools and Platforms: AI/LLMs, MS Excel, Microsoft 365, Google Workspace, Microsoft Power Platform

EMPLOYMENT

Federated Hermes Inc., Pittsburgh, PA

May 2025 - August 2025

Data Engineering Intern

- Authored notebooks in Databricks using Python and SQL that are still used by the Enterprise Data Platform team today
- Developed a checkpointing system for sequential overnight notebook workflows, enabling recovery from failures at any step and reducing full reruns by preserving prior successful outputs
- Rapidly resolved critical data routing errors to maintain uninterrupted company operations in emergency situations
- Designed and completed an individual capstone project that analysed and visualized Databricks notebook runtime beyond what already existed to allow further understanding of where optimizations could be made

Technical Lead for Team StarSync, Intern Hackathon

- Served as technical lead for team of interns on Hackathon project StarSync; designed an app using Microsoft Power BI that utilized a custom-trained AI chatbot to assist company employees in their own personal stock trade requests

Math Tutor, Pittsburgh, PA

May 2024 - August 2024

- Coached and mentored students in middle/high school level math, SAT/ACT prep, and college math preparation

LEADERSHIP

Cohon University Center, Carnegie Mellon University, Pittsburgh, PA

February 2023 - May 2026

Supervisor (Sept 2023 - May 2026); Information Desk Attendant (Feb 2023 - Aug 2023)

- Supervise main campus building and student desk attendants; oversee evening events and independently resolve any issues regarding facilities, equipment, scheduling conflicts; attend to emergencies by notifying appropriate personnel
- Respond to requests and answer questions from campus visitors, faculty and staff; manage university lost and found

PROJECTS

Danish Recycling Team Project, Artificial Neural Networks and Deep Learning, DIS

April 2025

- Created and trained a convolutional neural network to better identify the three different standardized types of recyclable plastic bottles in Denmark through image recognition

GPS Data Analysis Project, Numerical Methods, Carnegie Mellon University

November 2024

- Developed program in Python using nonlinear least squares algorithms such as Gauss-Newton to solve GPS positioning problems and analyze system conditioning for multiple satellite configurations

Risk112 Project, Fundamentals of Programming and Computer Science, Carnegie Mellon University

May 2023

- Coded in Python an online version of the board game "Risk" using object-oriented programming, top down design, and various data structures including recursive back-tracking and pixel manipulation

INTERESTS: Violin, Piano, Skiing, Travel, NFL, NBA, Geography, History, Astronomy